

RETNIFY

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Problem Overview

Goal: Accurately classify OCT (Optical Coherence Tomography) images into four categories: CNV, DME, DRUSEN, and NORMAL.

Why it matters: Early detection of retinal diseases is crucial for preventing vision loss. Automating classification can help make diagnosis faster and more consistent.

Project Objective: Train and evaluate deep learning models to identify the most effective solution with high generalization performance across unseen data.



Dataset Summary

- **Classes:**
 - CNV (Choroidal Neovascularization)
 - DME (Diabetic Macular Edema)
 - DRUSEN
 - NORMAL
- **Data Format:** Grayscale OCT images
- **Preprocessing:**
 - Resize all images
 - Applied weighted sampling to address class imbalance
 - Normalized pixel values for consistency

Summary of EDA Insights

Content:

- **Class Imbalance:**

- CNV and NORMAL were overrepresented
- DRUSEN and DME were underrepresented

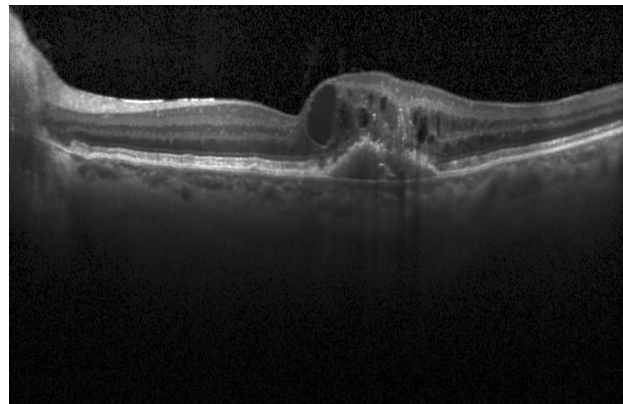
- **Image Quality:**

- Images varied in size - resized 224x224

- **Insight:**

- Needed class balancing for fair model training
- No major outliers or corrupted images

	Class	Train Count	Test Count
0	CNV	37205	250
1	DME	11348	250
2	DRUSEN	8616	250
3	NORMAL	26315	250





Final Modeling Pipeline

- Image Preprocessing
 - Resize all images to 224x224 pixels to match ResNet input requirements
- Data Augmentation
 - Applied only to the training set to improve generalization
 - Random Horizontal Flip
 - Random Rotation
 - Normalization and resizing applied to all sets
- Addressing Class Imbalances
 - Calculated class weights based on inverse frequency
 - WeightedRandomSampler for balanced batch sampling
 - Used weighted CrossEntropyLoss during training



Final Modeling Pipeline

- Model Architecture: ResNet-50
- Integrated SE block (Squeeze-And-Excitation)
 - Helped model focus on relevant retinal patterns
- Training Configuration
 - Trained for 50 epochs using Adam optimizer
 - Learning rate of 0.0001
- Tracked metrics: Loss, Accuracy, F1-Score (both train & validation)



Key Experiments and Metrics

Architecture Comparison

- Tested SimpleCNN, VGG-16, ResNet-18 and ResNet-50 + SE blocks under identical training regimes

Validation vs. Train Accuracy

- SimpleCNN: 92.15 % (val) / 93.84 % (train)
- VGG-16: 97.08 % / 99.61 % → large over-fit gap
- ResNet-18: 96.96 % / 99.19 %
- **ResNet-50 + SE: 97.21 % / 99.06 %** → smallest gap

F1-Scores (Validation)

- SimpleCNN: 0.9238
- VGG-16: 0.9706
- ResNet-18: 0.9696
- **ResNet-50 + SE: 0.9722**

Test-set Summary (1000 images)

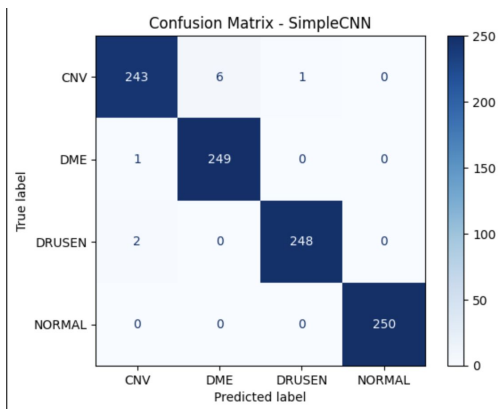
- ResNet-50 + SE: 97.2 % accuracy, 0.97 F1
- Only 2 misclassifications (both DME ↔ DRUSEN)

Interpretability & Robustness

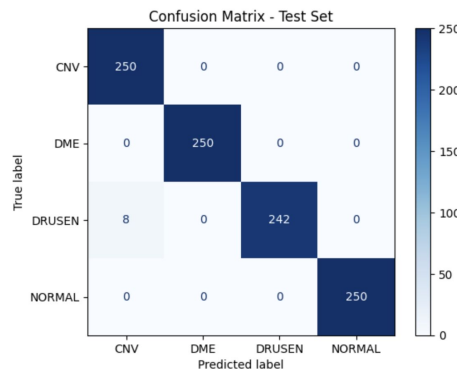
- Grad-CAM confirms focus on fluid pockets and disrupted layers

Key Experiments and Metrics

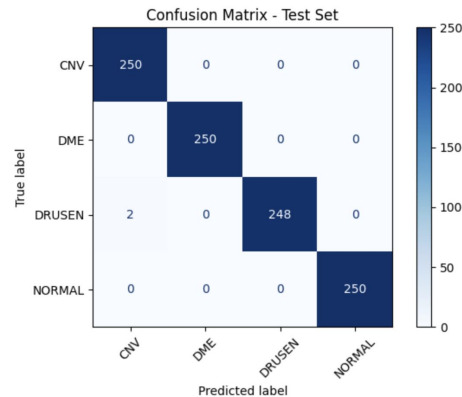
SimpleCNN



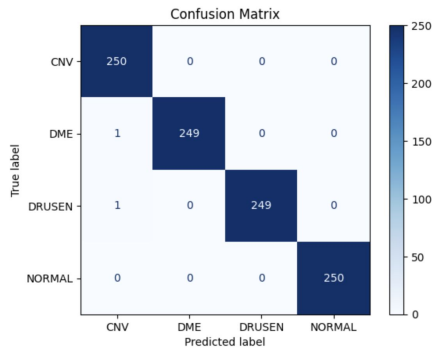
VGG16



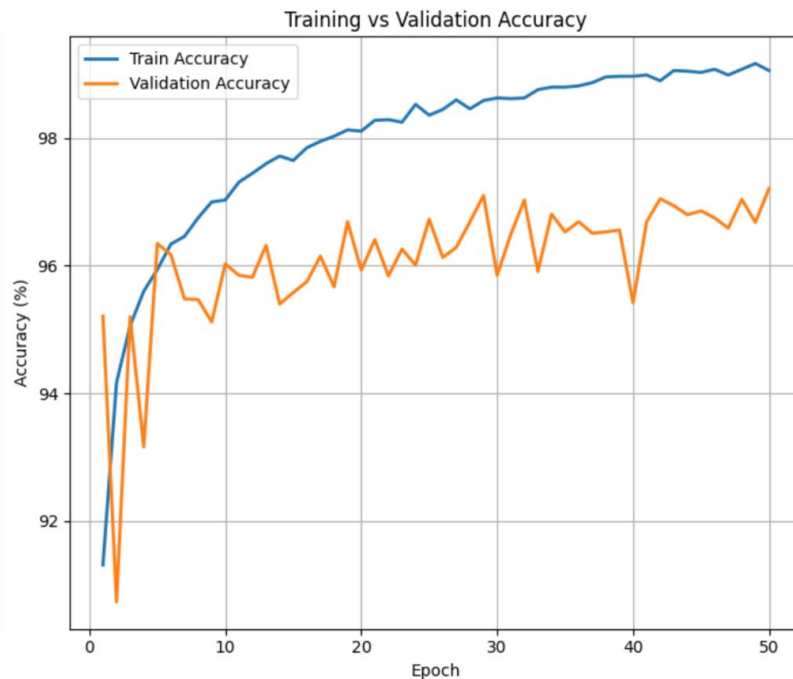
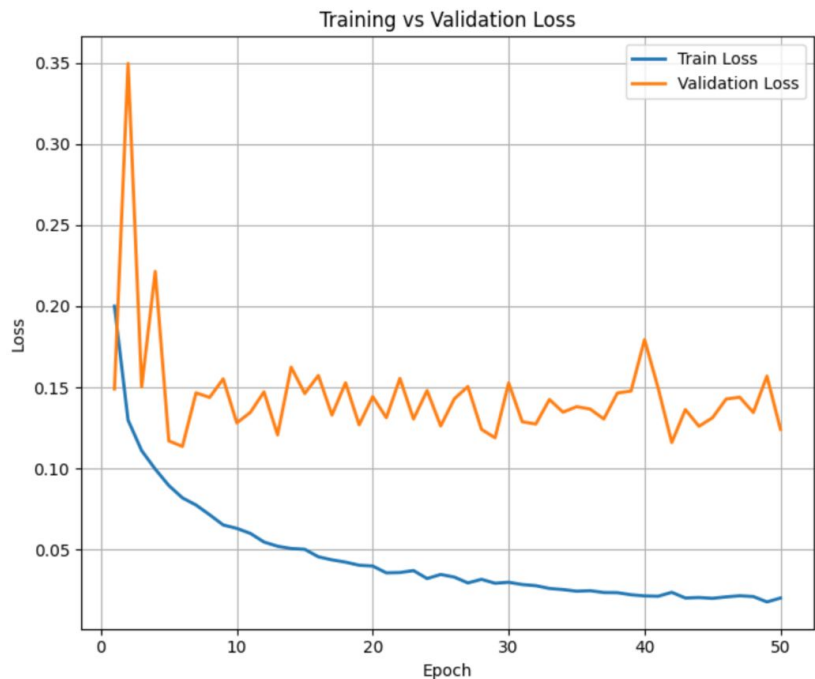
ResNet-18



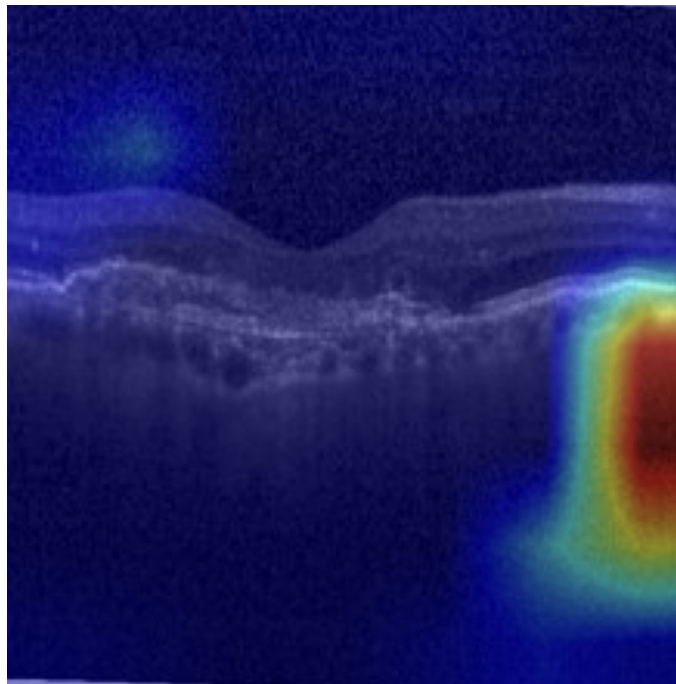
ResNet-50



Key Experiments and Metrics



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What Worked, What Did Not

- Weighted Random Sampler for Class Imbalance
- Data Preprocessing
- Long Training Times



Reflections & Contributions

- Baseline Model #1: SimpleCNN - Gaurav
- Baseline Mode #2: VGG16 - Idil
- Refined Model: ResNet-18 - Wu
- Final Refined Model: ResNet-50 - EJ

THANK YOU!

